Quiz 1 Solutions: Applied Natural Language Processing
DATA 441/641

Choose exactly 10 questions to answer!

1. What is stemming in NLP?
   (a) A method that relies on dictionary-based transformations to obtain base word forms
   (b) A process that produces grammatically correct words from inflected forms
   (c) A supervised learning approach for text normalization
   (d) **A technique for reducing words to their root form using predefined linguistic rules**

2. What is contrastive learning?
   (a) A reinforcement learning strategy where contrastive feedback helps improve predictions
   (b) A method that directly assigns class labels to unstructured data based on cosine similarity
   (c) A clustering technique that groups similar data points based on predefined labels
   (d) **A method that optimizes a model by maximizing the similarity between augmented views of the same data while minimizing similarity between different data points**

3. How does Part-of-Speech tagging contribute to syntactic analysis?
   (a) By extracting named entities to identify people, organizations, and locations
   (b) By analyzing sentiment polarity in textual data
   (c) **By classifying words into grammatical categories, enabling better sentence structure analysis**
   (d) By converting words into their base forms to standardize vocabulary

4. Why is lemmatization an important NLP preprocessing step?
   (a) **It converts words to their dictionary form to normalize textual data**
   (b) It extracts key phrases from a text for improved topic modeling
   (c) It assigns sentiment labels to words to support classification tasks
   (d) It is a dictionary-based process that removes any words that do not contribute to the overall meaning

5. What is stacking in machine learning?
   (a) A hierarchical model selection approach that optimally assigns nonlinear weights to various learning algorithms
   (b) **A method where a meta-model learns from the outputs of base models**
   (c) A sequential process of layering models, where each model learns from the errors of the previous one
   (d) A technique where multiple models are trained separately, and their predictions are averaged for better performance

6. Which NLP technique is used to identify persons, organizations, or locations?
   (a) POS
   (b) **NER**
   (c) Normalization
   (d) Sentiment Analysis

7. What is pretraining, and why is it fundamental in deep learning?
   (a) A mechanism to introduce noise into the training process to enhance robustness
   (b) A training strategy that solely relies on labeled datasets from the beginning
   (c) **A multi-phase learning approach where a model is exposed to vast amounts of data before being specialized for a task**
   (d) A simple initialization method that randomly assigns weights to a neural network

8. What is random augmentation in deep learning, and how does it support self-supervised learning?
   (a) A supervised learning approach that increases dataset size through manually annotated additional data
   (b) A method where new feature representations are learned dynamically during training through manual intervention
   (c) **A strategy that applies stochastic transformations to create varied versions of data, enabling loss computation without labeled samples**
   (d) A technique where generative adversarial networks synthesize additional training examples to expand datasets

9. What is a pretext task in self-supervised learning?
   (a) A rule-based approach for optimizing dataset quality before model training
   (b) A supervised learning step where a pre-trained model is fine-tuned for a downstream task
   (c) **An auxiliary task designed to help a model learn meaningful representations from raw data without explicit labels**
   (d) A post-processing step applied to improve feature selection in supervised learning

10. What is fine-tuning in deep learning, and why is it critical for domain adaptation?
   (a) A method where models are designed to ignore prior learning to prevent overfitting
   (b) A complete retraining of a neural network from scratch using a new dataset
   (c) **A technique where a pretrained model is adjusted using a smaller dataset to better capture the specific nuances of a target domain**
   (d) A process where only the final output layer is trained while all previous layers remain frozen

11. What is the role of negative sampling in contrastive learning?
   (a) A clustering method that partitions data into multiple disjoint groups
   (b) A technique for removing noisy labels from the dataset
   (c) A hyperparameter tuning strategy used to improve classification accuracy
   (d) **Selecting dissimilar data points to enhance representation learning by reinforcing meaningful feature separation**